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# From vision to export success: examining how and when entrepreneurial leadership drives export innovativeness

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**Abstract**

**Purpose** – Our paper examines how entrepreneurial leadership (EL) impacts firm export innovativeness via digital transformation and supply chain agility. Using the dynamic capability view theory, our paper also explores the moderating impact of leaders' boundary-spanning behavior on the connection between EL and export innovativeness.
**Design/methodology/approach** – To gauge the study hypotheses, time-lagged data were gathered from 244 firms in China. Confirmatory factor analysis was performed to assess the psychometric properties of the measures. The hypothesized links were gauged via the PROCESS macro.
**Findings** – Digital transformation and supply chain agility serially mediate the impact of EL on firm export innovativeness. Leaders' boundary-spanning behavior strengthens the positive effect of EL on digital transformation and supply chain agility, enhancing the drivers of export innovativeness.
**Practical implications** – Our study highlights the significance of boosting EL that supports digital innovation and agile supply chains. Encouraging leaders to actively engage across organizational and external boundaries can strengthen these strategic capabilities, ultimately driving export performance and innovation in the global markets.
**Originality/value** – Our paper makes a novel contribution by integrating leader boundary-spanning behavior into the connection between EL and firm export innovativeness, a field that has been largely neglected in the pertinent literature. Our paper advances the understanding of how EL fosters export innovativeness through key organizational capabilities, and how this process is shaped by individual-level boundary-spanning behaviors. The framework offers both theoretical insights and practical guidance for firms seeking effective market entry and long-term competitiveness in international markets.

**Keywords** Digital transformation, Entrepreneurial leadership, Firm export innovativeness, Leader boundary-spanning behavior, Supply chain agility

**Paper type** Research article

## 1. Introduction

In today's competitive marketplace, digital transformation enables firms to not only foster operational efficiency but also foster a culture of continuous improvement, learning, and adaptability that is imperative for sustainable growth and success in the long run (Khalid *et al.*, 2024). Through digital transformation, firms can optimize their business processes, improve decision-making, and capture market opportunities to remain competitive globally (Feliciano-Cestero *et al.*, 2023; Ivančić *et al.*, 2019; Schwertner, 2017). Thus, firms tend to invest in

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innovative activities to remain competitive and access global markets and opportunities (Freeman *et al.*, 2006; Wang *et al.*, 2020). Export innovativeness, which denotes "... the extent to which a firm develops novel product innovations targeted at export markets" (Boso *et al.*, 2013, p. 64), can play a vital role in this regard (Edeh *et al.*, 2020).

Export innovativeness is elucidated by the firm's ability to produce new products that are competitive, innovative, and radically different from competitors in a global setting. It enables firms to be more responsive to market opportunities (Lages *et al.*, 2009), enhance customer loyalty and satisfaction (Kunz *et al.*, 2011), and make the supply chain more resilient (Gölgeci and Ponomarov, 2015). Nevertheless, despite the argued importance of firm innovativeness, studies on firm export innovativeness are scarce (Boso *et al.*, 2013; Makri *et al.*, 2017). Thus far it seems that no empirical research has been employed on how entrepreneurial leadership (EL), digital transformation, and supply chain agility contribute to the development of firms' export innovativeness and market entry decisions in an international context. Additionally, little is known about how leadership styles and organizational level antecedents can facilitate businesses in improving firm export innovativeness in the global marketplace.

Despite the growing attention to export performance in international business research, limited empirical evidence exists on how firms develop the capabilities necessary to boost export innovativeness (Boso *et al.*, 2013; Edeh *et al.*, 2020). Although the strategic importance of export innovativeness is well-documented (Knight and Kim, 2009; Lages *et al.*, 2009), how and under what conditions EL enhances this capability in the global markets is not yet clearly understood. This gap is critical, as firms increasingly face dynamic, competitive, and technologically driven global environments that demand both adaptive leadership and innovation-centric strategies (Makri *et al.*, 2017; Varadarajan *et al.*, 2022).

To address said gap and respond to calls for research on export innovativeness (e.g. Verhoef *et al.*, 2021), our study draws upon the dynamic capability view (DCV) (Teece *et al.*, 1997) to examine the mechanisms through which EL enables firms to foster their export innovativeness. The DCV posits that in rapidly evolving environments, firms gain and sustain competitive advantage not through static resources, but via dynamic capabilities (Barreto, 2010; Teece *et al.*, 1997). Accordingly, we conceptualize EL as a firm-level capability that contributes to sensing opportunities, seizing them through innovation, and transforming resources. Leaders who exhibit entrepreneurial behaviors are more likely to stimulate organizational learning, allocate resources creatively, and drive initiatives that align with evolving global market demands (Asad *et al.*, 2021; Harrison *et al.*, 2018).

In view of the information presented above, we introduce digital transformation and supply chain agility as serial mediators linking EL to export innovativeness. These constructs are selected based on their relevance to dynamic capabilities in today's digitally enabled and supply chain–driven international markets. While the justification for their inclusion is rooted in both theoretical and empirical grounds, we articulate their roles in a focused and concise manner below.

Digital transformation denotes "... a process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies" (Vial, 2021, p. 118). In line with the DCV, digital transformation represents a sensing and seizing capability. It allows firms to ferret out emerging trends, boost customer experiences, and experiment with new business models (Fitzgerald *et al.*, 2014; Husnain *et al.*, 2025). Entrepreneurial leaders, through their innovation-oriented vision and willingness to challenge the status quo, are more likely to promote and implement digital transformation strategies that facilitate the global expansion (Khalid *et al.*, 2024). These strategies improve productivity, reduce operational inefficiencies, and facilitate firms to better adapt to changing global demands (Westerman *et al.*, 2014). Digital tools help firms strengthen their partnerships and coordination across the supply chain, lowering market entry barriers and expanding their global reach (Sahoo *et al.*, 2024; Schallmo *et al.*, 2017).

Supply chain agility reflects a firm's ability to swiftly adapt its supply chain operations in reaction to market volatility, technological shifts, and emergent opportunities (Gligor, 2014). It embodies the "transform" dimension of dynamic capabilities by allowing firms to reconfigure operations and build flexible logistics systems. Research shows that agile supply chains enhance responsiveness, reduce costs, and support competitive positioning in export markets (Ayoub and Abdallah, 2019; Gligor *et al.*, 2013). EL promotes agility by fostering cross-functional collaboration, open communication, and fast decision-making (Agarwal *et al.*, 2007; Kim *et al.*, 2017). Taken together, digital transformation and supply chain agility form a complementary pathway through which EL fosters firm export innovativeness.

Finally, we argue that the strength of this leadership–capability–performance linkage is conditioned by leader boundary-spanning behavior (LBSB). Boundary-spanning demonstrates the actions undertaken by leaders to build and maintain external relationships, acquire information and resources, and facilitate inter-organizational collaboration (Marrone *et al.*, 2007; Salem *et al.*, 2018). Leaders with high boundary-spanning behaviors engage in external scanning, resource mobilization, and stakeholder engagement, enabling them to access timely information and market intelligence that support digital transformation initiatives and agile supply chain configurations (Liu *et al.*, 2018; Van Meerkerk *et al.*, 2018). Moreover, such leaders activate team creativity, knowledge integration, and coordination across functional and national boundaries (Kim *et al.*, 2022), thereby reinforcing the positive effect of EL on firm capabilities and export innovativeness.

Our research contributes to the following different literature streams. Specifically, it delineates insights into research on the firm's new entry decisions and strategies in the international marketing context though past writings have highlighted that customer relationship management initiation, marketing information and insights, learning, flexibility, and strategic planning contribute to the firm innovativeness (Hult *et al.*, 2004; Nawaser and Jahanshahi, 2018). Our study highlights that EL can have a pivotal role in the firm export innovativeness through digital transformation and supply chain agility. With this realization, our paper augments current knowledge on the leadership-related predictors of firm export innovativeness that have centered upon internal marketing competencies and skills-related predictors of firm innovativeness so far (Hult *et al.*, 2004; Nawaser and Jahanshahi, 2018; Radas and Božić, 2009).

Research illustrates that EL gives rise to desirable consequences such as worker creativity and firm growth (e.g. Cai *et al.*, 2019; Harrison *et al.*, 2018). Considering these findings, we propose that EL engenders firm export innovativeness directly and via digital transformation and supply chain agility in the global markets. Broadly speaking, our paper assesses the role of digital transformation and supply chain agility as serial mediators of the influence of EL on firm export innovativeness. As such, our research enhances the literature on digital transformation (Matt *et al.*, 2015; Nadkarni and Prügl, 2021) as well as supply chain agility (Ayoub and Abdallah, 2019; Gligor, 2014; Gligor *et al.*, 2013). Finally, our study signifies the role of LBSB in gauging the influence of EL for digital transformation, supply chain agility, and firm export innovativeness in the global markets. Studies indicate that LBSB has been linked to desirable outcomes, such as creativity and work performance (Kim *et al.*, 2022; Liu *et al.*, 2018). By denoting LBSB as a boundary condition of the EL and digital transformation link, our paper enhances the pertinent literature. Figure 1 depicts the research model.

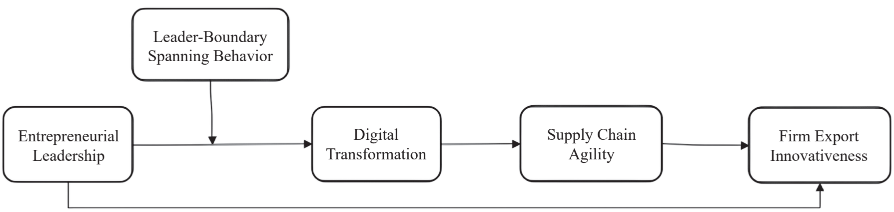


**Figure 1.** Research model. Source: Authors' own work

## 2. Theoretical framework and hypotheses

### *2.1 Dynamic capability view*

The DVC, which is a critical theory in dynamic settings and promotes adaptability, uninterrupted progress, and responsiveness to change (Teece *et al.*, 1997), is utilized to explain the proposed hypotheses. The DVC considers the challenges and uncertainties in today's competitive and modern marketplace, making it a wider, contemporary, and flexible approach for obtaining a competitive edge and accomplishing corporate growth in active markets (Bowman and Ambrosini, 2003; Wu, 2010). It highlights how crucial it is to possess knowledge, expertise, and skills that can change a firm's present resources, practices, and business operations (Ye *et al.*, 2022). It is important that businesses have the capacity to innovate and adapt in response to the internal and external challenges and changing environment (Gupta *et al.*, 2020). Dynamic capabilities surface from the business practices and actions that integrate, progress, and reorganize internal and external expertise and skills to swiftly address changing business settings (Vogel and Güttel, 2013). According to Wu (2010), dynamic capabilities are firms' problem-solving skills resulting from their tendency to see opportunities and challenges, make timely decisions based on market conditions, and continuously improve their resource base.

Dynamic capability comprises three elements: "sensing", "seizing", and "reconfiguring" (Teece, 2007). A firm's ability to investigate and take notice of opportunities both inside and outside of its boundaries and across its whole business value network and ecosystem denotes sensing (Teece, 2007). It entails understanding customers' needs and firm requirements, investing in research and development and technology and being informed of the innovations produced by business partners (Teece, 2007). Businesses regularly seek business possibilities within and outside of their network to obtain a competitive edge and drive growth (Gupta *et al.*, 2020). Seizing occurs once the organization has sensed the opportunity. It is defined as realizing the opportunity's value (Teece, 2007). It comprises establishing firm boundaries and taking proactive measures to capitalize on new trends, changes in the market, and evolving commerce needs (Teece, 2007).

Reconfiguring includes changing, restructuring, and rearranging its capabilities, resources, and organizational structure in response to changes in the marketplace (Teece, 2007). Evidently, reconfiguration is required to guarantee that the firm efficiently takes advantage of these opportunities recognized and mitigates the risks sensed (Vogel and Güttel, 2013). To summarize, the DCV highlights the importance of being aware of opportunities, taking proactive steps to seize them, and modifying the organizational structure or procedures to fully utilize the specified business potential.

### *2.2 Entrepreneurial leadership and firm export innovativeness*

EL figures out business possibilities, promotes new ideas, and makes organizational members get involved in innovative activities to enlarge and upgrade business (Fontana and Musa, 2017), which ultimately supports new market entry decisions. It is important to ascertain opportunities, devise actions to make use of those opportunities, adapt, and make the organizational practices in harmony with the changing environment and business potentials to reach success in the long run (cf. Khan *et al.*, 2023; Teece *et al.*, 1997). In essence, we surmise that taking possession of business expansion and growth opportunities is a critical part of EL. Firms are inclined to venture risks and have better capabilities to make creative and entrepreneurial choices that ultimately reduce entry barriers to international markets. Indeed, EL guides the firm on how to target foreign markets and integrate innovation in the business processes and products and improve marketing and selling tactics to be successful on a global scale. It ensures that the firm continuously innovates marketing offerings and processes to acquire a competitive advantage and market benefits in the global context.

Entrepreneurial leaders are innovative thinkers, creative, risk takers, have entrepreneurial skills, display a customer-focused approach, and possess long-term vision (Bani-Melhem *et al.*, 2025). Such leaders constantly canalize subordinates through their actions that make them endeavor towards shared goals and reach measurable growth through new market entry (Koryak *et al.*, 2015). EL delegates authority to their subordinates to be creative and sense novel ideas and opportunities via the enhancement of a culture of innovation, creativity, and risk tolerance (Bagheri, 2017), which is important to succeed and enter global markets. Entrepreneurial leaders encourage and efficiently distribute the resources to design new and creative products, improve product quality, and make them more cost-efficient (Asad *et al.*, 2024; Chen, 2007), which in turn assists them in achieving the market position and sustained growth in the global market.

EL's market-oriented decision-making and risk-tolerant mindset (Bussgang and Benbarak, 2016) reduce barriers to global entry by facilitating adaptive and forward-looking strategies. Unlike traditional leadership approaches, entrepreneurial leaders focus on mobilizing internal capabilities and orchestrating external resources, thereby positioning the firm to respond effectively to dynamic and competitive global environments (Leitch and Volery, 2017). Entrepreneurial leaders foster continuous learning, collaborative partnerships, and resource flexibility, which are essential for building export-oriented innovations tailored to the diverse and shifting demands of global markets (Chen, 2007). They encourage the development of integrated distribution networks and the acquisition of global customer insights, further enhancing a firm's ability to innovate for export markets. Based on these arguments, it is hypothesized that:

*H1*. EL relates positively to firm export innovativeness.

### *2.3 Entrepreneurial leadership and digital transformation*

Digital transformation has the potential to bring changes in society and industries using digital technologies (Agarwal *et al.*, 2010; Majchrzak *et al.*, 2016). At the organizational level, firms can find ways to innovate and improve their processes, operations, products, services, and overall performance of the business (Sahoo *et al.*, 2023; Vial, 2021). Studies suggest that digital technologies foster sustainable and long-term growth and market adaptability, identify new paths for value creation, and streamline the entire value chain (Cosa, 2024; Vial, 2021; Watini *et al.*, 2022).

The DCV places emphasis on the significance of identifying and seizing opportunities as well as rearranging resources to adapt effectively (Teece *et al.*, 1997). EL has a fundamental role in promoting innovation, learning, research and product development as well as driving and achieving economic and sustained growth (Koryak *et al.*, 2015; Leitch and Volery, 2017). In essence, EL contains a peculiar combination of "creativity", "adaptability", "proactiveness", and "risk-taking behaviors" (Bani-Belhem *et al.*, 2025; Renko *et al.*, 2015) that ultimately lead to successful market entry and operational success in foreign markets. EL frequently takes innovative positions (Zhan *et al.*, 2022), which are the driving forces to transform business processes and practices through emerging technologies. Their visionary approach may make it possible to integrate and utilize emerging digital technologies to search out market opportunities and boost processes and expertise that encourage firm export innovativeness.

Entrepreneurial leaders bolster a culture of innovation and help transform fresh perspectives into tangible enterprises for the purpose of responding to needs and demands in the marketplace (Kearney, 2020). They motivate subordinates to adopt digital technologies that improve business processes, customer engagement, product development, and overall efficiency, reduce overall operational costs, and advance overall efficiency, which is so important for the firm export innovativeness. They can get deeper insights and real-time data about foreign markets, customer preferences, and competitive forces through digital transformation for better product adaptations and target market strategies for each market, ultimately leading to successful market entry in the international market. Indeed, digital transformation may present few significant benefits that can enable firms to uncover new opportunities, skills, knowledge, and expertise (Ivančić *et al.*, 2019; Schwertner, 2017). Thus, such leaders may sense and capture these opportunities and concentrate on initiatives aimed at integrating and embracing digital technologies. As such, we posit that EL may realize the growing significance of digital transformation and invest in the cultivation of these skills that are needed for long-term success and access to the global market. This prompts the hypothesis that:

*H2.* EL relates positively to digital transformation.

### *2.4 Digital transformation and supply chain agility*

Digital transformation encompasses the adoption and integration of advanced digital tools and technologies that allow firms to identify emerging and diverse needs, providing firms with the ability to better forecast demand and customer preferences, market opportunities, and trends (Khalid *et al.*, 2025; Opazo-Basáez *et al.*, 2023) that enable firms to make a successful entry into the global markets. Consistent with the DCV (Teece, 2007), digital transformation can assist the entire supply chain to take advantage of new opportunities, develop novel solutions, improve experiences, and better predict customer demands, economic changes, and global trends in response to the changing market landscape.

Prior studies suggest that digital transformation alters the firm decision-making approach and finds innovative ways to create superior value through a deeper understanding of customers' and business partners' activities, processes, and approaches (Husnain *et al.*, 2025; Vial, 2021). Additionally, technological capabilities enable firms to better coordinate their business activities with their partners to accomplish sustainable growth and long-term success in the global markets (Eslami *et al.*, 2024). Thus, the supply chain mechanism becomes more agile, responsive, and flexible in response to changing business needs and global market opportunities.

Importantly, digital transformation improves visibility, resource efficiency, and decision-making throughout the entire supply chain (Al Mashalah *et al.*, 2022), enabling firms to develop operational flexibility, which in turn mitigates the barrier to the global market entry. Digital transformation helps firms reconfigure resources, alter production schedules, and adjust their logistics network in response to change or disruption (Dolgui *et al.*, 2020), giving rise to the development of innovative and competitive market offerings and solutions that are essential to gaining a competitive position and achieving sustained growth in the global context. Furthermore, digital transformation enables firms to collaborate with business partners (Khalid *et al.*, 2024), which finally promotes supply chain agility. It encourages firms to continuously acquire useful and valuable insights about suppliers and customers that help them to make supply chain and operational activities more agile and compete in the global markets. Accordingly, it is hypothesized that:

*H3.* Digital transformation relates positively to supply chain agility.

### *2.5 Supply chain agility and firm export innovativeness*

Teece (2007) states that organizational capabilities and skills are essential for efficiently integrating, managing, and exploiting resources to thrive in volatile business environments. In view of the DCV, we conceptualize that supply chain agility is deemed as a key resource to attain a competitive position and succeed in the global markets, as it assists firms to swiftly and effectively respond by reconfiguring, acquiring, and optimizing the resources to meet the emerging market needs. Supply chain agility makes all business activities from procurement to production and distribution more efficient, flexible, and responsive (Patel and Sambasivan, 2022), which in turn mitigates operational inefficiencies and enables them to be more competitive in the global markets. Indeed, supply chain agility implementation is key to achieving competitiveness and growth (Sangari *et al.*, 2015). It facilitates firms to be more responsive to the market needs and demands by fostering a culture of collaboration and production throughout the supply chain (Patel and Sambasivan, 2022) that finally provides a platform to access global markets. Agility enables the timely delivery of high-quality items to the right customer (Agarwal *et al.*, 2007). This is essential for enhancing customer satisfaction and customer overall experience, which in turn contributes to firm export innovativeness in the global markets.

Supply chain agility is deemed a key success factor for the firm business performance, as it allows firms to produce products with reduced lead times (Eckstein *et al.*, 2015; Tse *et al.*, 2016), ultimately reducing barriers to the global market entry. It enables firms to sense, perceive, and respond quickly to the changes in the business environment (Gligor *et al.*, 2016) by finding new and innovative ways of creating and delivering value to customers in a global context. With little disruption, an agile supply chain can adjust to changes in market opportunities, consumer demands, seasonal patterns, or financial situations. Through an agile supply chain, firms can develop more adaptive and customer-centric plans, contributing to firm export innovativeness. Thus, we argue that agility influences several supply chain activities, including sourcing, production, and distribution processes, ultimately contributing to firm export innovativeness in the global context. Thus, we propose the following hypothesis.

*H4.* Supply chain agility relates positively to firm export innovativeness.

### *2.6 Digital transformation and supply chain agility as serial mediators*

As postulated by the DCV, digital transformation and supply chain agility are critical tools to obtain a competitive advantage and stay ahead of the competitors (Dolgui *et al.*, 2020; Gligor *et al.*, 2016) in the global markets since they make firms acquire, integrate, improve, transform, and adjust their internal and external resources in reaction to customer demands, market trends, and disruptions. Digital transformation and supply chain agility both contribute to streamlining and optimizing the firm processes, resources, skills, knowledge, and activities (Al Mashalah *et al.*, 2022), which are key to entering global markets. As such, digital transformation offers valuable and deeper insights into business operations, global market trends, competitive forces, and customer preferences. These are important for strategic planning, resource allocation, and capturing potential market opportunities (Khalid *et al.*, 2024), while supply chain agility enables firms to respond quickly and effectively to changing business needs, market demands, and customer preferences (Gligor *et al.*, 2016; Tse *et al.*, 2016).

Both digital transformation and supply chain agility are essential to achieving a competitive position and making successful entry in export markets. In essence, EL facilitates the acquisition, development, integration, and improvement of key technological capabilities and skills (i.e. digital transformation), which are important to respond to the changing market landscape. Digital transformation in turn drives supply chain agility that ultimately activates firm export innovativeness in the global markets. In short, we surmise that digital transformation and supply chain agility act as serial mediators of the impact of EL on firm export innovativeness. Hence, it is hypothesized that:

*H5.* Digital transformation and supply chain agility serially mediate the effect of EL on firm export innovativeness.

### *2.7 Leader boundary-spanning behavior as a moderator*

The DCV posits that firms continuously update their resources in response to changes in the internal and external environment, identify opportunities, and develop strategies promptly to capitalize on those opportunities (Teece *et al.*, 1997). Businesses with a clear focus on understanding and serving customer needs can better recognize and avail themselves of those market opportunities (Khalid *et al.*, 2024; Usman *et al.*, 2024a) by reconfiguring the organizational resources, ultimately contributing to growth, expansion, and access to global markets. Essentially, the relationship of EL to digital transformation, supply chain agility, and firm export innovativeness is complex, and EL may depend on several factors. In this paper, we surmise that EL is subject to LBSB.

As proposed by the DCV, LBSB can assist firms in acquiring key resources and developing relationships with the key stakeholders to capitalize on market opportunities, which in turn reduce barriers to the global markets. Indeed, boundary-spanning behavior focuses on identifying risks, building resources, improving dialogs, and striving for knowledge on key issues (Salem *et al.*, 2018; Usman *et al.*, 2024b) and ensuring that digital transformation is aligned with the organizational long-term goals of success, growth, and new entry to global markets. Importantly, boundary-spanning behavior assists firms in understanding the importance of growth and expansion. Thus, leaders acquire, integrate, adjust, and utilize resources wisely to support the digital transformation process and ensure that integrating digital technologies can effectively contribute to the firm's export innovativeness in the global market.

In addition, individuals with more focus on boundary-spanning behavior try to acquire all the key resources and build strong relationships and networks. Consequently, they have more knowledge and are responsive (Salem *et al.*, 2018) to the emerging market needs, customer demands, and preferences, ultimately leading to firm export innovativeness compared to those who are low on boundary-spanning behavior. Given more emphasis on capturing market opportunities, building a strong network, market adaptation, and the ability to acquire and effectively use resources, individuals with high boundary-spanning behavior may make more efforts to integrate digital tools. Consequently, they engage in diverse activities and practices that help them to adopt and leverage digital technologies. To attain a competitive market position and access to global markets, LBSB can help firms to reach a better understanding of the organizational need for digital transformation. Thus, we posit that compared to others, individuals with high boundary-spanning behavior are likely to get involved in activities, procedures, and practices related to digital transformation. Therefore, it is hypothesized that:

*H6.* Leader boundary-spanning behavior moderates the positive effect of EL on digital transformation, such that this effect is stronger among leaders with high boundary-spanning behaviors than leaders with low boundary-spanning behaviors.

In light of the DCV, LBSB can advance business dynamic skills and transform business processes and practices, resulting in the achievement of competitiveness and access to the global markets. It can enable firms to acquire, develop, and allocate resources to digital transformation that positively contributes to achieving firm export innovativeness. They can efficiently control resources and prioritize organizational activities (e.g. fostering digital transformation and supply chain agility). Additionally, LBSB encourages digital transformation and fosters a culture that makes supply chain activities more agile, which in turn drives firm export innovativeness in a global context. In the present paper, we propose that the influence of EL on firm export innovativeness is through digital transformation and supply chain agility. Indeed, EL is deeply rooted in firms' planning efforts to access new markets through the appropriate usage of digital technologies (i.e. digital transformation) and supply chain agility, resulting in firm export innovativeness. Hence, it is hypothesized that:

*H7.* Leader boundary-spanning behavior moderates the positive indirect effect of EL on firm export innovativeness via digital transformation and supply chain agility, such that this effect is stronger among leaders with high boundary-spanning behaviors than leaders with low boundary-spanning behaviors.

## 3. Method

### *3.1 Sample and procedure*

To empirically test the proposed hypotheses, we adopted a time-lagged survey design, which enhances the reliability and internal validity of the study by diminishing potential common method bias. Data were collected from export-oriented firms operating in China, a context that offers a highly dynamic and globally integrated market environment. China's prominent role in international trade, coupled with its rapid digital transformation and evolving supply chain practices, provides a compelling setting to examine how EL influences export innovativeness. The exclusive focus on Chinese firms is theoretically and contextually justified. As one of the largest and most dynamic players in the global export market, China presents a fertile environment for investigating leadership-driven innovation. The institutional complexity, technological advancements, and international competitiveness of Chinese firms offer valuable insights into how EL, supported by digital and agile capabilities, contributes to export innovativeness in emerging markets.

Participants were drawn from the alumni network of a major public university in China. These individuals, occupying middle and senior management roles, were considered well-positioned to provide informed insights into their firms' leadership behaviors, digital practices, and export strategies. We initially contacted 800 alumni, out of which 540 consented to participate. Each respondent received a cover letter explaining the study's purpose, assuring strict confidentiality and anonymity.

Data were collected in four distinct time periods to allow for temporal separation among the independent, mediating, moderating, and dependent variables. At Time 1, participants completed a questionnaire measuring EL, LBSB, and demographic information. At Time 2, the focus was on digital transformation, while Time 3 measured supply chain agility. Finally, at Time 4, we collected data on firm export innovativeness. This sequential approach allowed for clearer causal inferences and reduced the likelihood of inflated correlations due to same-source bias.

Surveys were administered during working hours, and participants were encouraged to complete them independently to avoid external influence. To increase response rates, multiple reminders were sent through alumni employed within the participating organizations who also facilitated communication and logistical coordination. The number of surveys returned at Times 1 through 4 were 488, 423, 377, and 263, respectively. After screening the incomplete responses and failed attention checks, the final sample comprised 244 matched and usable responses. The response rate was 244/540 = 45.2%.

Given the reliance on self-reported measures, several steps were taken to minimize common method bias and enhance data credibility. First, the time-lagged design ensured separation between the measurement of predictor and outcome variables. Second, attention-check questions were included to ensure respondent attentiveness. Third, anonymity and confidentiality were assured. Additionally, we conducted Harman's single-factor test, which indicated that common method bias was not a serious concern. That is, the results from unrotated exploratory factor analysis denoted that the first factor accounted for the variance which was below 50%.

### *3.2 Measures*

All items were measured via a 5-point scale ("1 = strongly disagree" to "5 = strongly agree"). We measured EL with 8 items ($\alpha = 0.92$) (Renko *et al.*, 2015). An example item is "I often come up with radical improvement ideas for the products/services we are selling."

We measured digital transformation by adopting 5 items ($\alpha = 0.89$) from Nasiri *et al.* (2020). An example item is "We have created stronger networking between the different business processes with digital technologies."

We assessed supply chain agility with 5 items ($\alpha = 0.93$) from Blome *et al.* (2013). An example item is "We are able to adapt our services and/or products sufficiently fast to new customer requirements."

LBSB was gauged by adapting a 6-item scale ($\alpha = 0.88$) from Marrone *et al.* (2007). An example item is "I reach out to individuals outside of our organization who can offer expertise or ideas about the task at hand."

We gauged firm export innovativeness by using a 4-item scale ($\alpha = 0.91$) from Boso *et al.* (2013). A sample item is "Our company has produced more new products/services for our export markets than our key export market competitors during the past three years."

*3.3 Control variables*

To account for the potential confounding effects on firm export innovativeness, we included firm age and firm size as control variables (Boso *et al.*, 2013; Calantone *et al.*, 2006). In particular, firm size may affect the availability of resources for innovation and international expansion, while firm age may influence competitive dynamics and technological intensity, both of which are critical to export innovativeness. These variables were measured and statistically controlled to ensure the robustness of our findings.

## 4. Results

*4.1 Measurement model*

Confirmatory factor analyses were conducted in Mplus (8.8) to assess the measurement model that consisted of EL, LBSB, digital transformation, supply chain agility, and firm export innovativeness. All standardized loadings (Table 1) were significant ($p < 0.01$). The measurement model – $\chi^2(339) = 850.85$, $\chi^2/\text{df} = 2.51$, "Root mean square error of approximation", RMSEA = 0.07, "Standardized root mean square residual", SRMR = 0.06, "Comparative fit index", CFI = 0.90, "Tucker-Lewis fit index", TLI = 0.89 – fit the data satisfactorily. As shown in Table 1, coefficient alpha and composite reliability for each construct were > 0.70, providing evidence for the issue of reliability. Furthermore, all variables in Table 2 exhibited Average Variance Extracted (AVE) values surpassing 0.50. The square root of AVE for each variable was > its inter-construct correlations. Across all variables, both the Average Shared Variance (ASV) and Maximum Shared Variance (MSV) were < AVE, as outlined in Table 2. In short, convergent and discriminant validity were verified.

Table 2 presents the heterotrait-monotrait (HTMT) ratio results. Values below 0.85 indicate distinctiveness among the factors (Henseler *et al.*, 2015). As shown, all HTMT values were < this threshold, providing additional evidence for discriminant validity. Table 3 presented the summary statistics and correlations among the constructs.

*4.2 Hypotheses testing*

The hypothesized links were tested using the PROCESS macro. As depicted in Table 4, EL was positively associated with firm export innovativeness (hypothesis 1, $B = 0.44$, $p < 0.01$; 95% Confidence interval (CI) [0.28, 0.60]) and digital transformation (hypothesis 2, $B = 0.68$, $p < 0.01$; 95% CI [0.55, 0.81]). Thus, hypotheses 1 and 2 were corroborated. Digital transformation related positively to supply chain agility ($B = 0.29$, $p < 0.01$; 95% CI [0.16, 0.41]). Hence, hypothesis 3 was verified. Hypothesis 4 was also corroborated because supply chain agility related positively to firm export innovativeness ($B = 0.17$, $p < 0.01$; 95% CI [0.04, 0.30]).

The results revealed a significant indirect association between EL and firm export innovativeness via digital transformation and supply chain agility ($B = 0.03$, $p < 0.01$; 95% CI [0.008, 0.07]), indicating that digital transformation and supply chain agility serially mediated the positive association between EL and firm export innovativeness. Therefore, hypothesis 5 was supported.

We tested hypothesis 6 by including the interaction term involving LBSB and EL within the model. Based on the moderation analysis, as shown in Table 4, we found a significant positive effect of the interaction on digital transformation ($B = 0.13$, $p < 0.01$; 95% CI [0.06, 0.21]). Figure 2 demonstrated the exact characteristics of this moderated pathway. Simple slope plots

**Table 1.** Confirmatory factor analysis and internal consistency reliability

| Constructs | Items | Standardized loadings | Cronbach's alpha | Composite reliability |
|---|---|---|---|---|
| Entrepreneurial leadership | EL1 | 0.68 | 0.92 | 0.93 |
| | EL2 | 0.80 | | |
| | EL3 | 0.84 | | |
| | EL4 | 0.82 | | |
| | EL5 | 0.84 | | |
| | EL6 | 0.82 | | |
| | EL7 | 0.73 | | |
| | EL8 | 0.68 | | |
| LBSB | LBSB1 | 0.65 | 0.88 | 0.88 |
| | LBSB2 | 0.78 | | |
| | LBSB3 | 0.69 | | |
| | LBSB4 | 0.81 | | |
| | LBSB5 | 0.63 | | |
| | LBSB6 | 0.90 | | |
| Digital transformation | DT1 | 0.82 | 0.89 | 0.89 |
| | DT2 | 0.78 | | |
| | DT3 | 0.77 | | |
| | DT4 | 0.85 | | |
| | DT5 | 0.68 | | |
| Supply chain agility | SCA1 | 0.83 | 0.93 | 0.93 |
| | SCA2 | 0.86 | | |
| | SCA3 | 0.88 | | |
| | SCA4 | 0.87 | | |
| | SCA5 | 0.85 | | |
| FEI | FEI1 | 0.87 | 0.91 | 0.91 |
| | FEI2 | 0.82 | | |
| | FEI3 | 0.81 | | |
| | FEI4 | 0.88 | | |

**Note(s):** LBSB = leader boundary-spanning behavior; FEI = firm export innovativeness
**Source(s):** Authors' own work

**Table 2.** Validity tests and Heterotrait-Monotrait ratio

| Variables | 1 | 2 | 3 | 4 | 5 | AVE | MSV |
|---|---|---|---|---|---|---|---|
| 1. Entrepreneurial leadership | | | | | | 0.61 | 0.44 |
| 2. LBSB | 0.58 | | | | | 0.56 | 0.34 |
| 3. Digital transformation | 0.62 | 0.46 | | | | 0.61 | 0.45 |
| 4. Supply chain agility | 0.60 | 0.23 | 0.56 | | | 0.74 | 0.35 |
| 5. FEI | 0.66 | 0.52 | 0.66 | 0.55 | | 0.72 | 0.45 |

**Note(s):** FEI = firm export innovativeness; LBSB = leader boundary-spanning behavior; AVE = average variance extracted; MSV = maximum shared variance
**Source(s):** Authors' own work

displayed two specific conditional values of this moderated relationship: low LBSB and high LBSB. The link between EL and digital transformation was significant ($B = 0.38$, $p > 0.05$; 95% CI [0.20, 0.57]) when LBSB was low, while the link was significant ($B = 0.72$, $p < 0.01$; 95% CI [0.56, 0.88]) when LBSB was high. Therefore, hypothesis 6 was supported.

Additionally, the results of the moderation analysis indicated that the index of moderated mediation was significant for the hypothesized indirect relationship (via digital transformation

**Table 3.** Summary statistics and correlations of observed variables

| | Mean | SD | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|---|
| 1. Entrepreneurial leadership | 4.71 | 1.10 | – | | | | | | |
| 2. LBSB | 4.83 | 1.34 | 0.56** | – | | | | | |
| 3. Digital transformation | 4.55 | 1.32 | 0.55** | 0.51** | – | | | | |
| 4. Supply chain agility | 4.53 | 1.47 | 0.59** | 0.59** | 0.51** | – | | | |
| 5. FEI | 4.73 | 1.26 | 0.52** | 0.41** | 0.21** | 0.46** | – | | |
| 6. Firm age | – | – | 0.05 | −0.01 | −0.08 | 0.15* | 0.05 | – | |
| 7. Firm size | – | – | −0.03 | 0.03 | −0.06 | −0.05 | −0.06 | 0.14* | – |

**Note(s):** $N = 244$; SD = standard deviation; FEI = firm export innovativeness. LBSB = leader boundary-spanning behavior. *$p < 0.05$, **$p < 0.01$

**Source(s):** Authors' own work

**Table 4.** Hypotheses results

| | B | SE | 95% CI LL | UL |
|---|---|---|---|---|
| *Direct Paths* | | | | |
| Entrepreneurial leadership →FEI | 0.44** | 0.08 | 0.28 | 0.60 |
| Entrepreneurial leadership → Digital transformation | 0.68** | 0.07 | 0.55 | 0.81 |
| Digital transformation → Supply chain agility | 0.29** | 0.06 | 0.16 | 0.41 |
| Supply chain agility → FEI | 0.17** | 0.07 | 0.04 | 0.30 |
| *Indirect Path* | | | | |
| Entrepreneurial leadership → Digital transformation → Supply chain agility → FEI | 0.03* | 0.02 | 0.008 | 0.07 |
| *Moderated Paths* | | | | |
| Entrepreneurial leadership * LBSB → Digital transformation | 0.13** | 0.04 | 0.06 | 0.21 |
| Entrepreneurial leadership → Digital transformation (on low LBSB) | 0.38** | 0.09 | 0.20 | 0.57 |
| Entrepreneurial leadership → Digital transformation (on high LBSB) | 0.72** | 0.08 | 0.56 | 0.88 |
| Entrepreneurial leadership → Digital transformation → Supply chain agility → FEI (on low LBSB) | 0.02* | 0.01 | 0.004 | 0.05 |
| Entrepreneurial leadership → Digital transformation → Supply chain agility → FEI (on high LBSB) | 0.04* | 0.02 | 0.008 | 0.08 |
| Index of moderated mediation (FEI) | 0.007* | 0.004 | 0.001 | 0.02 |

**Note(s):** $N = 244$. B = Unstandardized coefficient; SE = Standard error, Bootstrapping specified at 5,000 with 95% confidence interval; CI = Confidence interval; LL = lower limit; UL = Upper limit; FEI = firm export innovativeness; LBSB = leader boundary-spanning behavior. *$p < 0.05$, **$p < 0.01$

**Source(s):** Authors' own work

and supply chain agility) between entrepreneurial leadership and firm export innovativeness (index = 0.007, $p < 0.05$, 95% CI [0.001, 0.02]). The Johnson-Neyman (J-N) plot was used to illustrate the indirect effect conditioned by LBSB, as proposed in Hypothesis 7 (Figure 3). This approach was chosen because the J-N plot is considered the most effective way to represent moderated mediation hypotheses (Preacher *et al.*, 2007). In Figure 3, the middle solid black line represents the mediated effect. Along this line, the highest point indicates a strong conditional mediated effect ($B = 0.04$, $p < 0.05$; 95% CI [0.008, 0.08]), corresponding to high values of LBSB (6.0). Conversely, the lowest point of the solid line represents a weaker conditional mediated effect ($B = 0.02$, $p < 0.05$; 95% CI [0.004, 0.05]), corresponding to lower values of LBSB (3.5). The area delimited by dotted lines around the solid line represents the 95% confidence band, with the upper and lower dotted lines indicating the respective limits of this interval. Overall, the plot demonstrated how the indirect effect of EL on firm export

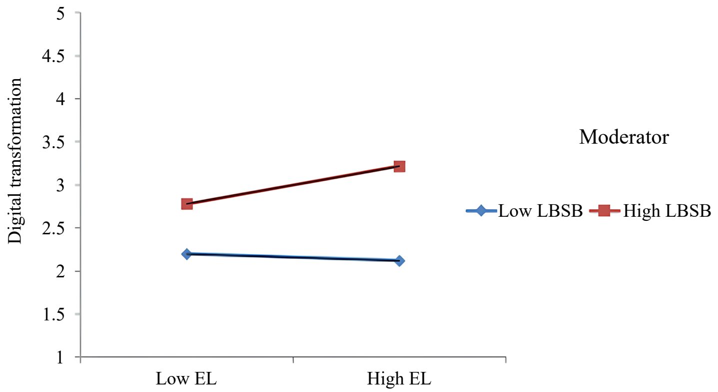


**Figure 2.** Entrepreneurial leadership and digital transformation: leader-boundary spanning behavior as moderator (hypothesis 6). Source: Authors' own work

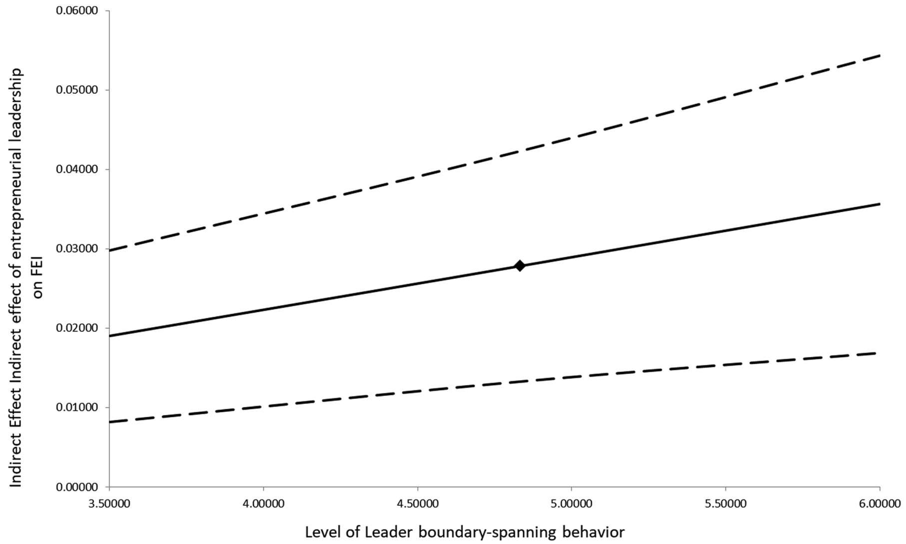


**Figure 3.** Indirect effect of entrepreneurial leadership on FEI through digital transformation and supply chain agility, for different levels of leader boundary-spanning behavior (hypothesis 7). Source: Authors' own work

innovativeness through digital transformation and supply chain agility increased as LBSB values rose, confirming hypothesis 7.

In closing, the inclusion of control variables did not result in any changes regarding the significance of the effects. For example, the indirect effect of EL on firm export innovativeness via digital transformation and supply chain agility was significant, and its indirect path (effect size) did not change.

## 5. Discussion

### *5.1 General findings*

Our paper examined: (a) the effect of EL on digital transformation and firm export innovativeness; (b) the impact of digital transformation on supply chain agility; (c) the influence of supply chain agility on firm export innovativeness; (d) digital transformation and supply chain agility as serial mediators of the impact of EL on firm export innovativeness; (e) LBSB as a moderator of the EL on digital transformation; and (f) LBSB as a moderator of the indirect impact EL on firm export innovativeness via digital transformation and supply chain agility. All hypotheses were confirmed, and the research model proposed in our paper was viable.

Drawing on the DCV (Teece *et al.*, 1997), we investigated digital transformation and supply chain agility as serial mediators, and LBSB as a moderator, to explain the mechanisms and conditions under which EL drove export innovativeness. Using a time-lagged design, the findings denoted that EL had a direct positive impact on export innovativeness, consistent with prior research emphasizing the role of leadership in driving innovation (Cardon *et al.*, 2009; Renko *et al.*, 2015). EL boosted digital transformation, while digital transformation bolstered supply chain agility, and supply chain agility stimulated firm export innovativeness. The findings demonstrated that EL enhanced export innovativeness indirectly through digital transformation and supply chain agility, which aligned with the DCV that firms should continuously adapt and reconfigure their capabilities to meet changing global demands.

The findings revealed that LBSB significantly moderated these relationships. Specifically, leaders who engage more actively in boundary-spanning activities amplify the impact of EL on digital transformation and export innovativeness. This finding supports previous studies suggesting that boundary-spanning facilitates knowledge acquisition, cross-border collaboration, and external key enablers of innovation and adaptability (Marrone, 2010; Salem *et al.*, 2018; Van Meerkerk *et al.*, 2018).

While these findings offer important contributions within the Chinese context, their broader relevance must be interpreted with caution. In economies with more advanced digital infrastructure and higher technological readiness such as Germany, Singapore, or South Korea, the incremental influence of EL on digital transformation may be less pronounced, given that such capabilities are already embedded within firm routines (Verhoef *et al.*, 2021). Conversely, in emerging or developing markets with limited digital maturity, EL may be even more critical in initiating and sustaining digital transformation efforts (Khalid *et al.*, 2024). Similarly, supply chain agility may vary substantially across global regions; in relatively stable environments, agility may serve as a competitive differentiator, while in volatile or fragmented supply chains as found in many parts of South Asia or Sub-Saharan Africa, agility becomes a fundamental survival mechanism (Ayoub and Abdallah, 2019; Gligor, 2014).

### *5.2 Theoretical implications*

Our study makes several important contributions to the pertinent literature. First, it extends the literature on the antecedents of firm export innovativeness by introducing EL as a critical and previously underexamined factor (Azar and Ciabuschi, 2017; Khalid *et al.*, 2024). While prior research has emphasized the roles of market orientation, organizational learning, and competitive intensity (Boso *et al.*, 2013; Makri *et al.*, 2017), our findings highlight that EL, which is designated by proactiveness, risk-taking, and innovation (Renko *et al.*, 2015), can drive firms' strategic decisions and behaviors toward entering new markets with innovative offerings. This shifts the focus from market- or firm-level antecedents to a more individual-level leadership construct, providing new insights into how export innovativeness is initiated and nurtured within firms.

Second, this research significantly contributes to the literature on EL in the global markets. While prior studies have linked EL to outcomes such as start-up growth (Ranjan, 2018), individual-, team-, and organizational level outcomes (Lin and Yi, 2025), adaptive innovation (Lin and Yi, 2023), and innovation (Newman *et al.*, 2018), our study is among the first to empirically examine its influence on export innovativeness in dynamic international settings. Our findings suggest that EL is instrumental in not only shaping innovation behavior but also in directing organizational strategies that enable firms to identify, exploit, and sustain competitive advantages in volatile global markets advancing current knowledge on leadership effectiveness in international business contexts.

Third, our study contributes to the digital transformation and supply chain agility literature by demonstrating their roles as serial mediators in the EL–export innovativeness relationship. While prior research has examined digital transformation in terms of improving organizational performance and public value (Bughin *et al.*, 2019; Mergel *et al.*, 2019), we extend this by showing that digital transformation is not just a technological shift but a dynamic capability, facilitated by leadership, that supports innovation and international expansion. Likewise, supply chain agility has traditionally been associated with operational responsiveness and resilience (Ayoub and Abdallah, 2019; Gligor *et al.*, 2016). Our findings position it as a strategic enabler of global market entry when driven by EL and digital capabilities.

Fourth, by integrating LBSB into our framework, our research advances the theoretical understanding of how individual-level differences interact with organizational processes to shape export outcomes. Our results suggest that leaders who display boundary-spanning behaviors by building external networks, acquiring market intelligence, and linking internal and external stakeholders can amplify the effect of EL on digital transformation and export innovativeness (Van Meerkerk and Edelenbos, 2018; Yan *et al.*, 2020; Zhu *et al.*, 2023). This integration offers a nuanced view of leadership effectiveness, highlighting how boundary-spanning behavior enables firms to better sense and seize global market opportunities, making a novel contribution to the literature on leadership in dynamic international environments.

### *5.3 Practical implications*

The findings offer several important implications for practitioners. First, the findings underscore the need for export-oriented firms to develop and support EL capabilities as a core strategic resource. Firms should actively identify and nurture leaders who demonstrate entrepreneurial traits such as opportunity recognition, innovation, and risk-taking. For example, cross-functional leadership training, international exposure programs, and innovation labs can cultivate the mindset and skills needed for global market success.

Second, structured development programs associated with EL should be embedded into organizational culture and strategy. Leaders should be empowered to form agile, cross-departmental teams including personnel from research and development, marketing, supply chain, and finance to co-create export strategies and drive innovation. Creating a risk-tolerant, learning-oriented environment enables these leaders to experiment, learn from failure, and continuously adapt to shifting global demands.

Third, while digital transformation and supply chain agility remain vital, our findings offer specific strategic guidance. Specifically, managers should align digital investments with market-sensing activities led by entrepreneurial leaders. Investing in advanced digital tools such as artificial intelligence, blockchain, and the Internet of things can support leaders in making timely, data-driven decisions that foster global competitiveness. However, the success of these technologies depends not only on their adoption but also on the vision and execution capabilities of entrepreneurial leaders.

Fourth, to realize the full potential of digital transformation, firms should establish a collaborative innovation culture that encourages partnerships with global stakeholders. Entrepreneurial leaders can play a key role in initiating strategic alliances, joint ventures, and co-creation initiatives with international customers and suppliers, thereby enhancing the firm's adaptability and reach in export markets.

Fifth, in terms of supply chain agility, firms should invest in technologies and processes that promote responsiveness to market shifts and disruptions. Entrepreneurial leaders should be equipped to design flexible supply chain architectures that can accommodate customization, reduce lead times, and minimize risk exposure, particularly in politically or environmentally unstable global regions.

Finally, our findings regarding LBSB have direct managerial relevance. Firms should assess and promote this capability as part of leadership evaluations and training. Leaders who actively engage with external stakeholders (e.g. regulators, suppliers, customers, and industry networks) are better positioned to ascertain emerging trends, anticipate regulatory shifts, and access strategic resources. As global markets become more complex and interconnected, boundary-spanning behavior will be critical in helping firms navigate uncertainty, manage external relationships, and deliver customer-centric innovations. Firms may also consider assigning boundary-spanning roles such as innovation ambassadors or partnership coordinators to boost market sensing and responsiveness in export initiatives.

*5.4 Limitations and future research*

While this study offers important contributions, it is not without limitations. First, although the use of a time-lagged design strengthens causal inferences by temporally separating the measurement of predictor, mediator, and outcome variables, it still falls short of establishing definitive causality. Future research should employ longitudinal or panel data designs to better capture how the relationships between EL, digital transformation, supply chain agility, and export innovativeness evolve over time, particularly in response to external shocks or market fluctuations.

Second, our study focuses exclusively on firms in China, which may limit the generalizability of the findings. Cultural norms around leadership, innovation, and risk-taking vary significantly across national contexts. For instance, hierarchical leadership cultures in some regions may inhibit entrepreneurial behaviors, while high digital readiness economies may reduce the relative importance of leadership in initiating transformation. Future research should replicate and extend this framework in diverse cultural and institutional environments, examining how cultural dimensions such as power distance, uncertainty avoidance, and individualism-collectivism (Hofstede, 2001) moderate the effectiveness of EL and related constructs.

Third, although the study identifies digital transformation and supply chain agility as key mediators, future work should explore alternative or additional mediating mechanisms that link EL to export innovativeness. Specifically, organizational learning, knowledge sharing, or dynamic market orientation could be integrated to provide a richer understanding of how firms convert leadership capabilities into export performance. The inclusion of psychological factors such as leaders' cognitive flexibility or emotional intelligence may help uncover further pathways through which leadership affects global strategic outcomes.

Fourth, while this research emphasizes LBSB as a key moderator, other contextual and environmental factors may also condition the effectiveness of EL. Future research should investigate the moderating roles of government policy, institutional support, industry turbulence, or global market volatility, particularly in post-pandemic economies where uncertainty and supply chain disruptions remain high. These external factors may shape the extent to which EL contributes to innovation and market expansion.

Fifth, our study focused on EL as the central leadership style influencing firm capabilities. However, other leadership approaches, such as strategic leadership, transformational leadership, and particularly digital leadership, may offer distinct pathways for enhancing digital and agile capabilities. Comparing these styles across contexts would yield important insights into which forms of leadership are most effective in driving innovation and international growth.

Lastly, future research should delve deeper into the technological dimension of global market entry. As emerging technologies such as blockchain, artificial intelligence, big data analytics, and advanced robotics become increasingly embedded in global operations, future studies could examine how these technologies interact with leadership and organizational practices to form firm innovativeness and export strategies. Doing so would offer insights into how technology-driven capabilities facilitate competitive advantage and long-term growth in international markets.

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